

Research paper

IoMT and explainable AI-enabled wearable system for classifying tremor and motor patterns in Parkinson's disease

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ABSTRACT

The prevalence of Parkinson's disease (PD) represents a significant global health concern due to its debilitating motor symptoms, particularly tremors. This pilot study presents an interpretable Internet of Medical Things (IoMT)-based monitoring system for analyzing hand tremors in individuals with PD. The proposed framework integrates a 6-axis inertial sensor for motion tracking, a low-power wireless microcontroller for data acquisition and connectivity, and a gradient boosting-based machine learning model (LightGBM) for tremor classification. During the data acquisition phase, signals from a triaxial accelerometer and gyroscope are sampled at 66.67 Hz, covering the characteristic 4–6 Hz Parkinsonian tremor frequency band. A RESTful API transmits the acquired data to a server over Wi-Fi, where it is stored in a relational database. Signal processing includes noise reduction, temporal segmentation, and frequency-domain feature extraction using the fast Fourier transform (FFT). The LightGBM classifier categorizes motor activity into Parkinsonian tremor, voluntary movement, or absence of tremor. Additionally, a web-based user interface supports real-time signal visualization and structured clinical data entry. As a feasibility study, the system demonstrates strong classification performance, achieving 95.64% accuracy and an F1-score of 0.95 on a dataset comprising 10,314 samples from nine participants (four with clinically diagnosed PD and five healthy controls). Although evaluated under controlled laboratory conditions, these results support the potential of the proposed system as a low-cost, interpretable tool for objective tremor assessment and continuous monitoring of Parkinson's disease symptoms.

1. Introduction

Parkinson's disease (PD) is a progressive disorder of the nervous system that disrupts motor control [1]. Symptoms develop gradually, with tremors (often starting in one hand) being the most common, along with muscle stiffness, slow movement, and balance impairments that increase fall risk. Symptom presentation varies across individuals, typically beginning on one side of the body and progressing bilaterally, though they usually remain more severe on the initially affected side [2].

In 2016, an estimated 6.1 million people worldwide were living with PD, representing the point prevalence for that year and marking a 145% increase since 1990 [3]. This trend is projected to continue in the coming decades [4]. By 2040, the World Health Organization predicts that neurodegenerative diseases such as PD and Alzheimer's will become the second leading cause of death globally, surpassing cancer-related mortality [5].

PD imposes a substantial economic burden due to its chronic and disabling course. Disorders of the central nervous system generate disproportionately high indirect costs, particularly in PD, since its typical onset coincides with a period when many patients and caregivers remain in the active workforce [6]. Beyond financial impact, individuals with PD face limitations in daily activities requiring fine motor control of the upper extremities, such as reaching and grasping [7].

Technological tools offer opportunities to enhance disease management and quality of life. Remote monitoring systems, for example, enable continuous home-based tracking of symptoms, providing objective data that complement subjective clinical evaluations and support more personalized treatment strategies [4]. Nonetheless, PD diagnosis remains challenging due to its clinical complexity and reliance on specialized expertise, often leading to inaccuracies. Thus, there is a pressing need for accessible, cost-effective, user-friendly, and reliable tools to improve detection and monitoring [8].

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In this context, Machine Learning (ML) has been widely explored to design intelligent systems capable of identifying characteristic PD patterns across diverse datasets [9]. By applying algorithms to large volumes of clinical data, ML uncovers meaningful patterns and supports computer-aided diagnostic systems, enabling the automatic classification of previously unseen cases [10].

To address these monitoring challenges, technological solutions have evolved rapidly. High-quality inertial measurement units (IMUs) have demonstrated that commercial systems, such as the PKG Watch and the Kinesia system, and consumer smartwatches, such as the Apple Watch, can perform reliably in clinical environments. Although effective, these commercial solutions have limited adoption in resource-scarce contexts, such as developing countries. This is due to their high cost and their operation as “black boxes”, where neither processed nor raw data, nor the algorithms created, are accessible.

This limitation has driven the growth of the Internet of Medical Things (IoMT), which integrates physical sensors with digital analytics to enable remote patient monitoring [9]. In this context, accessible technologies are essential. Regarding hardware, inertial sensors like the MPU6050 have become popular for diseases with motor symptoms due to their low power consumption and sensitivity to detect tremor frequencies between 4 and 6 Hz [11,12].

Regarding software, gradient-boosting frameworks such as LightGBM have become an attractive alternative to deep learning for portable applications. For example, LightGBM offers faster inference times and lower memory requirements, making it particularly well-suited for cloud edge computing environments [13].

Nevertheless, a notable gap persists in the existing literature. Numerous studies have proposed low-cost prototypes based on ESP32 microcontrollers (Espressif Systems) and MPU sensors (InvenSense) (e.g., [14–16]). However, these works primarily focus on data collection or basic classification accuracy. They often overlook two fundamental clinical requirements: (1) the ability to handle unbalanced data, given that tremor events do not occur continuously, and (2) interpretability. In contrast to previous low-cost approaches, our method incorporates Explainable Artificial Intelligence (XAI) techniques to explicitly support the interpretation of model predictions. This design choice directly addresses the pervasive “black box” trust issue associated with existing affordable monitoring prototypes.

Therefore, the primary objective of this pilot study is to design and validate a low-cost, open-architecture IoMT wearable system for classifying hand motor patterns in PD. Specifically, the secondary objectives are: (1) to classify Parkinsonian tremor versus voluntary movement and no tremor using a cost-effective IMU system; (2) to integrate SHAP-based interpretability to provide transparent explanations for model predictions; and (3) to demonstrate the feasibility of a low-cost end-to-end pipeline suitable for resource-constrained environments.

The main contributions of this pilot study are consolidated into a comprehensive framework. First, we present a portable, open-architecture IoMT prototype that leverages a generic 6-axis inertial sensor and a wireless microcontroller, offering a replicable, cost-effective alternative to closed-source commercial solutions. Second, the system uses the Synthetic Minority Oversampling Technique (SMOTE) to address challenges posed by small initial sample sizes and the natural scarcity of tremor events. Third, using explainable artificial intelligence (XAI) via SHapley Additive exPlanations (SHAP) to offer interpretable insights into model decisions. Finally, LightGBM is selected for its efficiency, making it suitable for wearable applications. This integrated approach represents a step toward accessible, interpretable monitoring tools for PD.

2. Related works

This section presents a comprehensive review of previous studies related to tremor monitoring in PD [17]. The first subsection reviews hardware-based approaches and analyzes the role of inertial sensors,

accelerometers, and other integrated systems in the objective analysis of tremors. The second subsection explores various tremor-monitoring systems, analyzing existing portable and non-portable technologies for tracking tremor severity and frequency. The third subsection delves into the applications of machine learning in PD, focusing on the algorithms used for tremor classification and patient assessment. Finally, articles linking explainable artificial intelligence to PD are reviewed.

2.1. Hardware design and integration

The selection of hardware, particularly sensors for data acquisition, is fundamental for PD monitoring systems. Among these, Inertial Measurement Units (IMUs) are predominant. Carissimo et al. [18] proposed a low-cost device that integrates an LSM6DSM IMU from STMicroelectronics. This system-in-package includes a 3D digital accelerometer and gyroscope, with a power consumption of 0.65 mA in high-performance mode [19]. This sensor was characterized in previous work by Addabbo et al. [20] with respect to its frequency response. Similarly, the work by Huo et al. [21] mentions a wearable system incorporating three IMUs, complemented by force and mechanomyography (MMG) sensors. In another approach, Carissimo et al. [14] developed an ad hoc network framework using Movella XSens wearable IMU sensors, demonstrating the feasibility of collecting motion data for PD monitoring. The versatility of IMU sensors is further evidenced by research work exploring their integration in different formats and locations. For example, Zhao et al. [22] used the Shimmer3 sensor as a wrist-worn device to collect data during the performance of Movement Disorders Society-sponsored Unified Parkinson's Disease Rating Scale (MDS-UPDRS) activities. In addition, devices such as the MPU6050, a MEMS sensor combining a 3-axis accelerometer and gyroscope, with an integrated Digital Motion Processor (DMP), have been used for the analysis of fall, gait, and tremor, as described in [15]. The integration of inertial sensors into everyday devices, such as smartwatches equipped with gyroscopes, has also been explored for quantifying motor symptoms in response to treatments such as deep brain stimulation [23]. Likewise, IoMT-based systems propose using sensors, including the MPU9250 (with a 3-axis accelerometer, a 3-axis gyroscope, and a magnetometer), for the specific study of hand tremors [16]. In addition to inertial sensors, other data acquisition modalities have been investigated. Ding et al. [24] presented a computer vision system using a camera for detecting Parkinson's gestures and tremor, demonstrating the feasibility of hand image analysis for this purpose. An innovative approach is presented by Xue et al. [25], who developed a wearable ultrasound (WUS) array with a center frequency of 7.5 MHz for real-time monitoring of muscle activity and tissue displacement analysis. As for microcontrollers, data processing and sensor data aggregation are crucial tasks. Carissimo et al. [18] employed a low-power 32-bit ARM microcontroller (STM32WB55), which integrates a floating-point unit (FPU), a real-time adaptive accelerator, and Bluetooth Low Energy (BLE) connectivity. Similarly, the framework proposed by Carissimo et al. [14] uses a Raspberry Pi 4 as a data-aggregation and communication device, enabled by the MQTT protocol. In developing a smart spoon for patients with PD, Singhal et al. [26] used a NodeMCU ESP-32S microcontroller to manage inertial sensor data and other components. The use of low-power microcontrollers is a key factor in the design of wearable devices for continuous and long-term monitoring of PD symptoms [18]. The selection and integration of these hardware components must consider factors such as cost, power consumption, patient ease of use, and the metrological characteristics of the sensors to ensure reliable and clinically relevant data acquisition for analysis and decision-making [18].

In summary, the literature demonstrates a mature ecosystem of low-cost hardware components (e.g., IMU sensors such as the MPU6050 and microcontrollers such as the ESP32) that are technically capable of capturing the kinematic signature of Parkinsonian tremor. However, this maturity in data acquisition has not systematically translated into comprehensive solutions. Existing prototypes often function primarily as technological proofs of concept, emphasizing data-capture

feasibility or raw classification accuracy, while remaining largely disconnected from clinical practice. Little discussion exists about how these systems address the inherent class imbalance in continuous monitoring data or how they translate algorithmic outputs into interpretable information for clinicians, thereby perpetuating the “black box” nature of even low-cost solutions.

2.2. Monitoring software

The development of robust, user-friendly software is crucial to the effective implementation of PD monitoring systems. This software must facilitate the connection and reception of data from various hardware devices, and provide an intuitive user interface (UI) for patients, caregivers, and healthcare professionals to visualize and analyze relevant information. Regarding connectivity and data transfer, various technologies and protocols have been explored. Carissimo et al. developed a sensor node based on an STM32WB55 microcontroller with Bluetooth Low Energy (BLE) connectivity to transmit preprocessed data to a cloud platform [18]. Similarly, Carissimo et al. proposed an ad hoc network framework using Movella DOT IMU sensors that communicate via BLE to a Raspberry Pi 4 acting as a data aggregator [14]. The Raspberry Pi 4 runs Python software that enables communication via the Message Queuing Telemetry Transport (MQTT) protocol. The choice of MQTT is based on its design for efficient data transmission over low-bandwidth connections, an essential feature in IoMT applications. Other research also mentions wireless data transmission. LeMoyné et al. [23] describe using a smartwatch application to record gyroscope signals and wirelessly transfer them to an email account, serving as a temporary cloud computing resource [23]. Dinesh et al. developed an IoMT-based glove that uses an ESP32 for wireless data collection, processing, and transmission via Bluetooth for automatic tremor assessment [16]. In analyzing fall, gait, and tremor, the MQTT protocol is used to continuously upload IMU sensor and load cell data to a broker for analysis and visualization in a desktop application [15]. With respect to software platforms and user interfaces, special attention has been paid to real-time data visualization and presentation of information in an accessible manner. Carissimo et al. [14] used the open-source IoMT platform ThingsBoard to develop a dashboard for visualizing statistics across monitored subjects and sensors. This dashboard includes graphs depicting the temporal evolution of relative energy in the resting and kinetic tremor frequency bands, a motor state indicator (ON/OFF), a pie chart categorizing tremor severity, and the temporal evolution of tremor severity occurrences. In addition, an alarm window is incorporated to monitor the patient's condition and set the control level when tremor increases or when medication is needed. The system proposed by Anusha et al. continuously uploads IMU sensor data to an MQTT broker for real-time display via a Serial plotter [15]. Their desktop application for fall, tremor, and gait analysis features user interface elements that indicate normal parameter states (tilt, weight, vibration) using green signals, along with visual and audible alerts upon event detection. A threshold configuration panel is also included to customize the detection sensitivity. Real-time display of tremor patterns is also an essential functionality. The adaptability and customizability of the software platforms are another relevant aspect. The framework proposed by Carissimo et al. [14]. Allows both the transmission of raw data from IMU sensors for processing and visualization by technical experts, and the delivery of aggregated and preprocessed data for rapid inspection of high-level features by medical specialists. This provides flexibility and adapts to the needs of different users. In short, PD monitoring software focuses on establishing reliable wireless connections to hardware devices (mainly IMUs) using protocols such as BLE and MQTT. Open-source IoT platforms, such as ThingsBoard, and custom desktop applications are used to create intuitive user interfaces that enable real-time visualization of relevant data, detection of significant events (such as alarms for increased tremor or changes in motor status), and customization of the information displayed to different user types. The trend is towards systems that facilitate continuous and remote

monitoring, providing healthcare professionals with tools for timely intervention and a better understanding of disease progression.

Conversely, while numerous low-cost, open-architecture prototypes exist, they frequently prioritize basic functionality and accuracy, overlooking two key clinical needs: robust handling of inherently imbalanced data where tremor events are sparse, and the provision of interpretable insights to clinicians. This work directly targets this gap.

2.3. Machine learning in Parkinson's disease

The application of ML algorithms has emerged as a fundamental tool for analyzing data to detect, classify, and monitor PD and its motor symptoms. These algorithms, trained on multimodal data from various sources, aim to identify complex patterns and nonlinear relationships to enable an objective, continuous assessment of the patient's condition. Neural networks (NN), both traditional and deep (DNN), have been extensively explored in this field. Carissimo et al. designed and tested two NN classifiers deployed on a microcontroller: one using information from the six degrees of freedom (DOF) of an IMU (accelerometer and gyroscope) and the other based solely on accelerometer data [18]. In the test phase, the 6 DOF NN achieved an accuracy of 99.59%, while the 3 DOF NN achieved 98.48% [18]. Jyotiyana et al. developed a PD classification model trained with a DNN, reporting an accuracy of 94.87% [27]. Sarankumar et al. also used deep neural networks, specifically to enable predicting PD severity [28]. Chen et al. used an ML framework with two-step feature selection and a generic elastic-net regularization-based model to evaluate PD using smartphone data [29]. Support Vector Machines (SVMs) represent another widely used ML algorithm. Huo et al. [21] employed a voting classification technique based on SVM, Multilayer Perceptron (MLP), AdaBoost, and k-nearest neighbors (k-NN) classifiers to predict Unified Parkinson's Disease Rating Scale (UPDRS) scores. In the study of leg tremors, various types of SVMs (linear, quadratic, cubic, Gaussian fine, Gaussian medium, and Gaussian coarse) demonstrated high classification accuracies [30]. Likewise, tree-based algorithms have also been shown to be effective. Huo et al. [21] included Decision Tree and AdaBoost in their voting classification approach. The gait analysis study reported in [31] indicates that decision trees and SVMs can achieve classification accuracies of up to 99.4% across PD stages. Gonçalves et al. [32] developed a gait monitoring system for PD patients, although the specific algorithm is not detailed in the excerpt provided. Other ML algorithms explored include k-Nearest Neighbors (k-NN), which was used by Huo et al. [21] and reported a classification accuracy of up to 99.64% for hand tremor detection in an IoMT-based glove [16]. Logistic Regression was applied to classify hypomimia in patients with Parkinson's disease [33]. Naive Bayes (NB) was evaluated as an alternative to k-NN and was employed in Parkinson's disease classification studies, particularly for tremor prediction using EEG data [34]. The LightGBM algorithm was used by Raj et al. [35] to classify PD stages based on features extracted from the visibility graphs of speech and gait signals, achieving an average accuracy of 85.3% in a multimodal configuration (speech and left-foot gait). LeMoyné and Mastroianni successfully applied a Multilayer Perceptron (MLP) [23] to evaluate the efficacy of deep brain stimulation for PD, achieving 90% accuracy. Finally, XGBoost was also explored experimentally as an alternative to LightGBM in the same study by Raj et al. [35]. In addition to traditional classification algorithms, Deep Learning techniques have received increasing attention. Chen [29] proposed an ML framework with potential applications in Deep Learning. The study by Alotaibi and Mahmood proposes a specialized deep convolutional neural network architecture for gait recognition that is robust to standard covariates, such as view angle and carrying conditions, and demonstrates competitive performance on the CASIA-B dataset without requiring data augmentation or fine-tuning [36]. San-Segundo et al. investigated PD tremor detection in uncontrolled environments using wearable accelerometers [37]. Sigcha et al. propose an automated system for assessing resting tremor in Parkinson's patients using smartwatch accelerometers and multitask convolutional

neural networks, demonstrating high accuracy and feasibility for continuous, non-intrusive monitoring [38]. The current trend is towards multimodal analysis, combining data from different sensors and sources to improve the robustness and accuracy of ML models. The selection and tuning of the hyperparameters of these algorithms, as well as rigorous validation using techniques such as cross-validation, are crucial steps to ensure generalization and reliability of the results in real clinical applications. Model interpretability, through techniques such as SHAP, is also gaining importance for understanding the contributions of different features to the diagnostic and monitoring process.

The landscape of ML applied to PD tremor is diverse and reports very high accuracy. However, a critical examination reveals two common trends: first, many studies use balanced or curated datasets that do not reflect the sporadic distribution of tremor in the real world; second, the predominant focus remains on the accuracy metric, treating interpretability as a secondary complement. This creates a disconnect between the reported performance under ideal conditions and practical utility in a continuous monitoring scenario, where clinician trust depends on understanding the “why” behind each prediction.

2.4. Explainable artificial intelligence

Ducange et al. [39] address the challenges of preserving privacy and model explainability in AI-based healthcare systems by integrating federated learning (FL) and XAI, referred to as Fed-XAI. Conducted in scenarios where sensitive patient data is distributed across multiple medical centers and cannot be centralized, the study evaluates trade-offs among model accuracy, interpretability, and transparency across heterogeneous data distributions. The results show that, while the neural network generally achieves higher accuracy, the fuzzy rule-based system offers competitive performance with superior interpretability, highlighting the potential of Fed-XAI to balance predictive performance with reliability in privacy-sensitive medical contexts. Similarly, Priyadharshini et al. [40] provide a comprehensive ML framework, enhanced with XAI, for the early diagnosis of PD using 3D magnetic resonance imaging scans from the PPMI database. The approach includes a preprocessing stage based on segmentation and radiomic feature extraction, followed by the selection of the 20 most statistically significant features through a two-step process validated using ANOVA and other statistical tests. A Gradient Boosting (GB) classifier, combined with SMOTE to address data imbalance, achieved 96.8% accuracy, outperforming other models. XAI techniques such as LIME, SHAP, and SHAPASH were integrated to reveal the contribution of specific GLCM texture features (e.g., difference entropy, joint entropy) to predictions, improving transparency and clinical confidence. The framework demonstrates great potential for accurate, interpretable PD diagnosis, though it remains computationally intensive and requires further real-world validation and user-friendly clinical integration. The application of various ML algorithms [41], including Decision Trees, XGBoost, and AdaBoost, for early PD prediction, with a strong emphasis on model transparency through XAI techniques. By employing feature importance analysis, the research identifies key variables that influence predictive outcomes, offering clinically relevant insights to support medical decision-making. The findings reveal that the AdaBoost classifier achieved the highest accuracy of 98.34%, demonstrating both its effectiveness and computational efficiency compared to more resource-intensive deep learning approaches, thereby making it suitable for real-time applications [42,43]. Beyond predictive performance, integrating XAI addresses the critical need for interpretability in AI-driven healthcare tools, fostering trust among practitioners and facilitating understanding of underlying decision processes [41]. The study thus bridges a gap in the literature by combining cost-effective ML models with explainability, ultimately aiming to enhance early PD detection, improve patient outcomes, and lay a foundation for future advancements in predictive modeling of neurodegenerative diseases.

Although XAI techniques like SHAP and LIME are being adopted in healthcare for high-dimensional data (e.g., MRI), their integration into

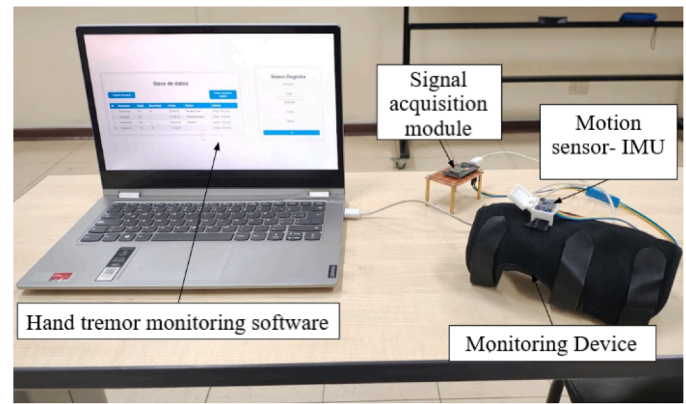


Fig. 1. Prototype of the tremor monitoring system showing the MPU6050 sensor connected to the ESP32 microcontroller.

the pipeline of low-cost, edge-deployable IoMT systems for motor symptom monitoring is absent. This leaves clinicians without insight into the reasons behind a device’s classification, undermining trust and clinical utility.

2.5. Identified research gap

In summary, while advancements in sensors, ML, and XAI are evident, a significant disconnect remains. The literature lacks a cohesive, low-cost IoMT framework that simultaneously: (1) employs affordable, accessible hardware; (2) explicitly addresses the class imbalance problem inherent in continuous tremor monitoring; (3) utilizes an efficient ML model suitable for edge deployment; and (4) embeds model interpretability as a fundamental feature to bridge the gap between algorithmic output and clinical insight. This study is designed to fill this specific gap by proposing and validating an integrated system that addresses all four aspects in unison.

3. Materials and methods

This section describes the materials and methods used to develop a tremor-monitoring system for PD. It covers: signal acquisition via the MPU6050 sensor and the ESP32 platform, including data sampling and calibration; signal processing techniques for feature extraction; data storage protocols; the application of ML algorithms for tremor characterization and explainability, with a user interface designed for data visualization and interaction.

3.1. Signal acquisition

The process of acquiring signals for the detection and analysis of PD tremors is detailed: the core components are the MPU6050 sensor and the ESP32 microcontroller, integrated for efficient data acquisition and wireless transmission (Fig. 1). The USB cable shown provides power during the experimental setup, while data transmission operates exclusively via Wi-Fi to ensure IoMT functionality.

3.1.1. MPU6050 sensor and ESP32 integration

The MPU6050, a 6-axis MotionTracking device combining a 3-axis gyroscope and a 3-axis accelerometer, was selected for its high precision and ability to capture detailed motion and orientation data [11]. This sensor is crucial for detecting subtle tremor movements. The ESP32 microcontroller was integrated to enable real-time data processing and wireless transmission via Wi-Fi.

The sensor was affixed to the patient’s wrist. While finger-mounted sensors may offer superior sensitivity for certain tremor types, the wrist

Table 1
Specifications of the MPU6050 sensor.

Sensor Specification	Value
Accelerometer Range	$\pm 2g, \pm 4g, \pm 8g, \pm 16g$
Gyroscope Range	$\pm 250, \pm 500, \pm 1000, \pm 2000$ °/s
ADC Resolution	16-bit per channel
Communication Interface	I2C (up to 400 kHz)

g: acceleration due to gravity (9.80665 m/s²). °/s: degrees per second. kHz: kilohertz.

Table 2
Parameters of data sampling.

Parameter	Value
Chosen Sampling Period	15 ms
Sampling Frequency	66.67 Hz
Required Nyquist Frequency	40 Hz

position was selected to mimic the form factor of commercially available smartwatches (e.g., Apple Watch, PKG), thereby maximizing patient comfort and the potential for future integration into daily life [12]. This configuration facilitates accurate documentation of tremor characteristics (Table 1).

The ESP32's wireless capabilities were used to transmit data in real time to a processing unit, enabling continuous monitoring and analysis. The ESP32 also performed preliminary data processing, such as basic filtering and averaging, to reduce noise and data volume before transmission.

3.1.2. Data sampling

To effectively capture Parkinsonian tremors, a careful selection of the data sampling rate is crucial. The typical frequency ranges of these tremors are reported to be 3–5 Hz. To avoid aliasing and accurately represent these frequencies, the Nyquist-Shannon sampling theorem requires that the sampling rate be at least twice the highest frequency of interest. The Nyquist-Shannon sampling theorem is mathematically expressed as:

$$f_s \geq 2 \cdot f_{\max} \quad (1)$$

where: f_s is the sampling frequency and $f_{\max}$ is the maximum frequency component in the signal. Given that we aim to analyze tremor frequencies up to 20 Hz to effectively capture the full spectrum of tremor types, including rest (4–6 Hz) and higher-frequency action tremors (6–15 Hz) reported in recent biomechanical studies [12], the Nyquist frequency is 40 Hz. Therefore, a sampling rate exceeding 40 Hz is required. As detailed in Table 2, a sampling frequency of 66.67 Hz was chosen. This sampling rate ensures we capture sufficient data points per cycle of the tremor frequencies, enabling effective signal processing and analysis without aliasing.

3.1.3. Data collection protocol

The data collection protocol was designed to obtain a complete set of information on tremor, in compliance with the ethical guidelines established in ruling No. 199-2024-CEICVt/PUCP, corresponding to project No. 258-2024, approved by the Office of Research Ethics and Scientific Integrity (OETIIC), Peru. A total of 22 participants were initially recruited. Thirteen people were excluded because they were under 60 years old, as Parkinson's disease predominantly affects adults over 59 years of age. The result was a final cohort of 9 participants: 5 without PD (control group) and 4 diagnosed with PD. While this sample size represents a pilot feasibility study, the segmentation of continuous signals into multiple time windows generated a sufficient number of instances ($N > 6000$) for statistical training of the ML models. All participants provided written informed consent before the assessments.

The MPU6050 sensor was firmly affixed to each participant's wrist with an ergonomic glove or armband, guaranteeing continuous contact

and reducing motion artifacts. Each assessment, lasting approximately 10 to 15 min, was performed after the motor examination segment of the MDS-UPDRS and encompassed three standardized tests: (1) postural tremor of the hands, with arms extended and palms oriented downward; (2) kinetic tremor, utilizing the finger-to-nose maneuver; and (3) amplitude of resting tremor, with hands positioned on armrests for a minimum of 60 s. To establish the ground truth for labeling, a protocol was defined in consultation with neurologists specializing in movement disorders, based on the standardized tasks of the MDS-UPDRS. This protocol was applied manually: during data collection, real-time visual inspection flags were recorded to document the presence or absence of tremor or anomalous movements during each specific task. Labels were then assigned by the principal researcher according to the following criteria. For PD patients, segments from the resting task (3) were classified as "Parkinsonian Tremor" when visual inspection confirmed tremor presence. For both cohorts, segments from the postural (1) and kinetic (2) tasks were labeled as "Voluntary Movement" or "No Tremor". For control subjects, segments from the resting task (3) where no movement was observed were categorized as "No Tremor". This approach sought to ensure consistent ground truth assignment by combining the protocol-defined task structure with immediate visual verification. Throughout the session, sensor data were continuously recorded to capture natural manifestations of tremor, transmitted in real time to a processing unit via the ESP32 wireless connection, allowing for immediate analysis.

3.1.4. Calibration

The MPU6050 sensor was statically calibrated before data collection to determine baseline zero-offset values for the accelerometer and gyroscope, thereby ensuring accuracy and dependability. The device was positioned on a flat, level surface to calculate the mean bias for each axis over 1000 samples, which was subsequently subtracted from subsequent measurements. Raw data from the I2C interface were converted into SI units utilizing specific coefficients: 16384.0 LSB/g and the standard gravitational constant 9.81 m/s² for acceleration, and 131.0 LSB/(°/s) along with a $\pi/180.0$ factor for converting rotational speed to radians per second. Although no dynamic calibration was conducted during acquisition, preprocessing, such as outlier removal, axis-wise standardization, and the computation of resultant acceleration vectors, was applied to minimize dynamic errors and positional biases, providing a robust basis for subsequent analysis.

3.1.5. Environmental considerations

To ensure the high quality and reliability of the data obtained, several environmental considerations were taken into account.

Concerning the surroundings, data collection occurred in a confined space (roughly 10 × 10 m) chosen to minimize external disturbances. To minimize potential artifacts, the experimental apparatus was segregated from other electrical devices and positioned away from sources of mechanical vibration, such as heavy machinery or automotive traffic. The monitoring system comprises the MPU6050 sensor encased in a portable, low-roughness housing. PLA+ served as a biocompatible material to reduce skin irritation and perspiration, thereby improving patient comfort during data collection, as corroborated by recent dermatological applications of 3D printed splints [44]. Furthermore, to ensure secure sensor placement and reduce motion artifacts, the following materials were used: Velcro on the dorsal side of the hand for gauntlet attachment and secure fixation, and medical-grade alcohol for skin disinfection before application. By adhering to these environmental considerations and following careful data-acquisition protocols, we aimed to obtain reliable, high-quality data for subsequent signal processing and analysis (Fig. 2).

3.1.6. System performance and reliability

The performance and reliability of the end-to-end IoMT system were evaluated during the pilot study. In the controlled laboratory environment, the system demonstrated stable operation with no observable data



Fig. 2. Controlled environment for data collection.

loss; all transmitted packets were successfully received and stored in the database, maintaining the consistent 66.67 Hz sampling rate. This high reliability is attributed to the stable Wi-Fi connection and robust error handling in the ESP32 firmware and the server API.

Power consumption, a critical factor for wearability, was estimated from the typical operating currents of its components: the ESP32 microcontroller in active Wi-Fi mode (230 mA), the MPU6050 sensor (3.8 mA), and supporting circuitry. For a standard 1000 mAh lithium-polymer battery, this yields an estimated continuous operation time of approximately 3–4 h. For extended monitoring, strategies like duty cycling would be implemented in a future deployment.

While formal latency benchmarks were not the primary focus of this validation study, the architecture is designed for low-latency operation. The data flow from sensor sampling to database storage involves minimal processing and direct network transmission, typically resulting in sub-second latency in a local network. The LightGBM model inference time is negligible (typically < 50 ms). Future work will include rigorous quantification of end-to-end latency under varied network conditions.

3.2. Signal processing

Signal processing is a crucial step in filtering out noise, extracting relevant features, and preparing the data for analysis. This section describes the specific signal processing steps applied to the MPU6050 sensor readings.

3.2.1. Preprocessing

Signal preprocessing is a crucial step to guarantee the quality of data used in ML models, especially in biomedical applications. This procedure addresses outlier detection and elimination, data standardization, and temporal normalization, ensuring the homogeneity and representativeness of the final dataset. In this work, signals captured by triaxial accelerometers at a sampling frequency of 66.67 Hz are processed to extract relevant parameters characterizing each recorded episode.

a) Duration adjustment, oversampling, and normalization: To ensure uniform signal length, each recording was set to 1 min, with a sampling interval of 15 ms and 3–5 repetitions per patient, depending on fatigue level. Recordings exceeding 60 s were truncated before the next repetition, while shorter recordings were extended via oversampling with replacement to preserve the original statistical properties. Non-overlapping 1 s sliding windows were utilized for feature extraction in each segment, ensuring uniformity and consistency in data

Table 3

Extracted signal feature parameters.

Feature Parameter	Column Name
Per Axis (x, y, z)	
Mean	<i>axis_mean</i>
Standard Deviation	<i>axis_std</i>
Median	<i>axis_median</i>
First Quartile (Q1)	<i>axis_q1</i>
Third Quartile (Q3)	<i>axis_q3</i>
Minimum	<i>axis_min</i>
Maximum	<i>axis_max</i>
Dominant Frequency 1	<i>axis_peak1_freq</i>
Magnitude at Dominant Frequency 1	<i>axis_peak1_amp</i>
Dominant Frequency 2	<i>axis_peak2_freq</i>
Magnitude at Dominant Frequency 2	<i>axis_peak2_amp</i>

processing. Finally, a Min-Max normalization was applied to scale all feature values to the [0,1] range, reducing the impact of varying measurement magnitudes and facilitating model convergence during training.

b) Feature extraction: Descriptive statistics summarizing the dynamic properties of each processed signal were extracted. These include the mean, standard deviation, median, first and third quartiles, minimum, and maximum of each axis (x, y, z). Finally, the two main frequencies and magnitudes of each axis were added to provide a more detailed description of the data distribution within each time window. These characteristics allow each episode to be represented as a vector in an attribute space, suitable for subsequent use in supervised learning models.

3.2.2. Frequency analysis

To extract additional features in the frequency domain, the Fast Fourier Transform (FFT) was applied:

$$X(f) = \sum_{n=0}^{N-1} x(n)e^{-j2\pi f n/N} \quad (2)$$

where: $X(f)$ is the complex frequency spectrum of the signal, $x(n)$ is the signal in the discrete-time domain, N is the total number of samples of the signal, f is the frequency index (in units of discrete frequency), n is the sample index in the time domain, j is the imaginary unit defined as $j = \sqrt{-1}$, and $e^{-j2\pi f n/N}$ is the complex exponential basis function representing a sinusoid at frequency f . Since the analysis focuses on frequencies below 40 Hz, a bandpass filter was implemented in the 0.5–20 Hz range, thus avoiding the DC component and possible distortions at low frequencies. The filter used was a fourth-order Butterworth, which provides a flat passband response and an attenuation slope of 24 dB/octave, defined by the recursive difference equation:

$$y[n] = \sum_{k=0}^M b_k x[n-k] - \sum_{k=1}^N a_k y[n-k], \quad (3)$$

where $x[n]$ is the input signal, $y[n]$ is the filtered signal, b_k and a_k are the filter coefficients, M is the order of the non-recursive part (FIR), and N is the order of the recursive part (IIR). In the specific case of a 4th-order bandpass filter, we have $M = N = 4$. The coefficients $\{b_k\}$ and $\{a_k\}$ depend on the selected cutoff frequencies (f_{low} , f_{high}) and on the filter design method used (e.g., Butterworth, Chebyshev, or Elliptic). These parameters determine the shape of the filter's frequency response. These features complement time-domain statistics by characterizing the signal's main oscillatory components. In summary, the most relevant outcome of this analysis was the detection of peaks in the frequency spectrum and the precise identification of the frequencies at which these peaks occur, enabling an accurate characterization of tremor patterns (Table 3).

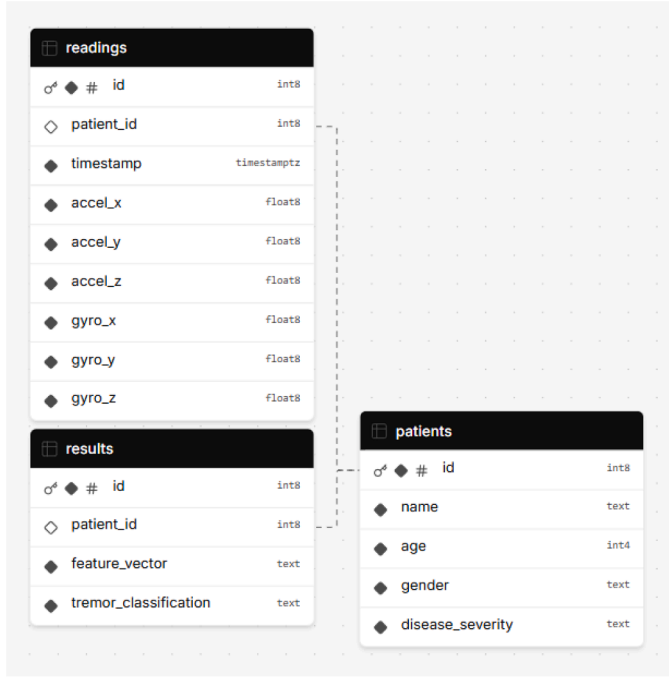


Fig. 3. Tables in the database.

3.3. Data architecture

Data storage is managed using a MySQL database designed for efficiently recording sensor readings and patient information.

3.3.1. Database structure

The database is structured into three main entities, interconnected by a foreign key that identifies each patient: Patient, Readings, and Results (Fig. 3). The first entity contains demographic and clinical information for each patient, including age, gender, and disease severity, in accordance with patient consent and data privacy regulations. The second entity stores all MPU6050 sensor readings, including timestamps, accelerometer data (x, y, z), and gyroscope data (x, y, z). A foreign key links each reading to the corresponding patient in the Patient entity. Finally, the third entity stores the results of the ML algorithm's tremor analysis and classification. It includes features extracted from the sensor data and the corresponding tremor classification results, also linked to the patient via a foreign key.

3.3.2. REST API and security

The ESP32 uses its Wi-Fi interface to connect to the server for data transmission. As an HTTP client, it transmits sensor data to the server in real time via POST requests. The REST API, developed with Node.js and Express.js, provides an endpoint for receiving and processing sensor data. Upon receipt of a POST request from the ESP32, the API validates the incoming data to ensure integrity and accuracy. Upon validation, the data are inserted into the Readings table of the MySQL database. Each data entry is linked to the corresponding patient through a foreign key referencing the Patient table. Additionally, the API provides endpoints to retrieve the most recent sensor readings for each patient, enabling real-time monitoring and analysis. The mysql2 library manages interactions with the MySQL database, facilitating efficient data storage and retrieval. To guarantee data security during this prototype phase, API access is limited using IP allowlisting and rudimentary authentication tokens. The design facilitates the implementation of HTTPS with TLS 1.3 encryption for the secure transmission of patient data, in compliance with medical data privacy regulations for future clinical deployment.

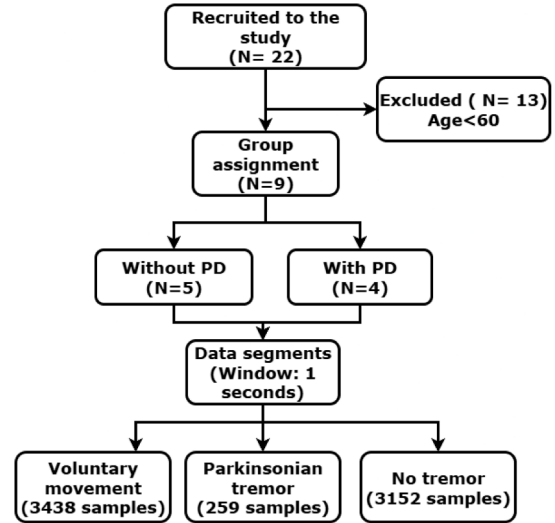


Fig. 4. Flowchart of participants and data segments of the second group.

3.3.3. MySQL database

The MySQL database serves as the persistent storage for sensor readings and patient information. In this implementation, the database is deployed locally and listens on port 8080, which facilitates development and testing in a controlled environment. For remote access and deployment in a production environment, a hosting service is required, with the database server configured for internet accessibility while implementing appropriate security measures. The database is organized into three main tables interconnected by foreign keys, enabling efficient data retrieval and analysis. This relational structure ensures data integrity and supports the execution of complex queries, providing a robust and scalable solution for storing and retrieving both sensor data and patient information.

3.4. ML algorithm application

The ML algorithm used in this study focused on classifying hand tremor patterns into three distinct categories: Parkinson's tremor, voluntary movement, and no tremor in healthy patients. This classification task is critical for differentiating pathological motor patterns, enabling more accurate and interpretable real-time monitoring of Parkinson's Disease (PD) symptoms.

3.4.1. Dataset and feature set

The dataset used in this study consisted of accelerometer recordings stored in CSV files, categorized into three target classes: voluntary movement (3438 samples), Parkinsonian tremor (259 samples), and no tremor (3152 samples). Data were collected from human participants, including both healthy individuals and patients diagnosed with Parkinson's disease. All participants performed standardized routines designed to elicit the three target movement patterns under controlled conditions (Fig. 4).

3.4.2. Model selection and training

Several ML algorithms were initially tested, including support vector machines (SVM), k-nearest neighbors (k-NN), Gradient Boosting (GB), Random Forest (RF), XGBoost, LightGBM, and CatBoost. The models were implemented using the scikit-learn library in Python, with hyperparameters optimized using Optuna.

To address the class imbalance (fewer Parkinsonian tremor samples) and ensure the generation of realistic synthetic patterns, we applied the Synthetic Minority Over-sampling Technique (SMOTE) strictly within the training folds. This prevents information leakage to the validation set. Given the limited number of subjects ($N = 9$), we employed a Group

K-Fold cross-validation strategy. This method ensures that all data segments from a single subject appear exclusively in either the training or test set for a given fold, preventing “subject leakage” in which the model learns the person’s identity rather than the disease pattern. The process was repeated across multiple runs, and model performance was assessed using accuracy, macro F1-score, macro precision, and macro recall, summarized as mean values with their corresponding standard deviations. The comparative outcomes of the models are presented in the Results section.

Optuna is a hyperparameter optimization framework that integrates efficient search methods, pruning techniques, and scalable distributed execution, enabling flexible, cost-effective optimization across a wide range of machine learning and non-machine learning tasks. Its dynamic, on-the-fly approach enables the construction of the search space in real time, providing significant performance improvements across benchmark tests and practical applications [45].

The LightGBM framework introduces two main techniques to accelerate Gradient Boosting Decision Trees (GBDT) without significantly compromising accuracy. Gradient-based One-Side Sampling (GOSS) retains all instances with large gradients and randomly samples those with small gradients, applying adjusted weights to maintain the quality of information gain. Exclusive Feature Bundling (EFB) reduces dimensionality by grouping mutually exclusive features using a greedy graph-coloring approach. Together, these methods enable LightGBM to train up to 20 times faster than conventional GBDT implementations while maintaining comparable accuracy [46].

3.4.3. Explainable AI integration

Lundberg and Lee present SHAP, a framework that unifies several existing model interpretation methods (e.g., LIME, DeepLIFT, Layer-Wise Relevance Propagation, Shapley regression/sampling) under the concept of additive feature attribution models, i.e., linear explanations in which each feature is assigned an importance value. By applying cooperative game theory, the authors demonstrate that only one method of this kind satisfies three desirable properties—local accuracy, data scarcity, and consistency—and that this method calculates Shapley values. SHAP values are defined as the average marginal contribution of each feature in all possible feature arrangements, interpreted as the change in the model’s expected prediction when that feature is conditioned [47]. To improve the model’s interpretability, the SHAP framework was integrated into the classification process. SHAP values were calculated for each feature in the dataset, providing information on each variable’s contribution to the final prediction.

3.4.4. Deployment and inference

Once trained and validated, the model was implemented using a Flask-based API, which interacts with the main web application and the database server. When new tremor data is received through the IoMT channel, it is preprocessed and sent to the classifier, which returns both the predicted tremor category and the corresponding SHAP explanation. This allows doctors or users to understand the reasoning behind the classification result transparently. The low computational complexity of the LightGBM model and the efficiency of the SHAP approximations ensured that classification and interpretation could be performed with minimal latency, making the system suitable for real-time or near-real-time deployment in clinical or home environments [48,49].

3.4.5. Implementation

The ML model was developed in Python with the scikit-learn library and deployed via a Flask server. For training, the dataset was used alongside cross-validation methods to ensure robust evaluation. In deployment, the trained model runs on a Flask server, enabling client applications to submit requests to classify new data entries.

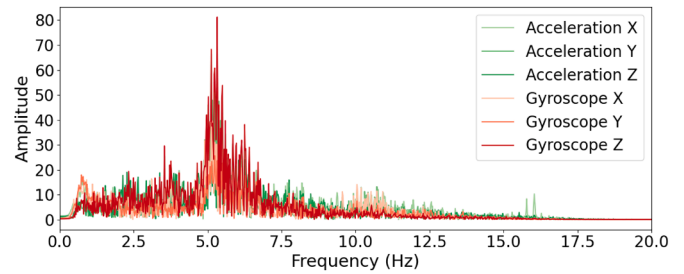


Fig. 5. Frequency Response of Filtered MPU6050 Signals (0.5–20 Hz) in a Parkinson’s Patient.

3.5. User interface

The user interface (UI) was developed as a web application that allows users to enter patient data and visualize tremor classification results in an accessible, intuitive manner.

3.5.1. Frontend

The frontend was implemented using HTML, CSS, and JavaScript to deliver an interactive, responsive experience. The application includes an entry form that allows new patients to be registered by entering their relevant data. Additionally, it provides a data display section where users can view real-time sensor readings alongside corresponding tremor classification results.

3.5.2. Backend

The backend was developed using Java Server Pages (JSP) with Apache Tomcat, enabling seamless integration between the user interface and the MySQL database. Using JSP, the system handles database queries for patient registration and for storing sensor readings. Furthermore, JSP integrates with a Flask server using the GSON library to perform requests and retrieve the classification results generated by the ML model.

4. Results

4.1. Dataset creation and validation

This subsection presents the procedures used to create and validate the dataset. The process included preprocessing of inertial signals, class balancing, and feature analysis, which are described in detail below.

4.1.1. Preprocessing steps

During the signal analysis phase, frequency-domain transformations were applied to data from both Parkinson’s and healthy subjects. The results revealed distinct frequency patterns associated with tremor activity. As shown in Fig. 5, the Parkinson’s patient exhibits a prominent peak at 5 Hz, which aligns with the typical frequency range of pathological tremor reported in the clinical literature. This frequency is a key indicator of tremor activity. In contrast, Fig. 6 shows the frequency response of a healthy subject, where the dominant frequency peak is typically below 3 Hz. This consistent emergence of a dominant spectral peak within the 4–6 Hz band for PD subjects, observed across all tremor-positive recordings, provided a quantifiable biomarker. It directly informed our feature extraction strategy, confirming the relevance of dominant frequency (*_peak1_freq) and magnitude (*_peak1_amp) features for the subsequent classification model.

The acceleration in Y (Fig. 7) exhibits well-defined periodic oscillations, with an estimated frequency between 4 and 6 Hz, corresponding to the characteristic range of resting tremor associated with Parkinson’s disease. Accelerations in X and Z also show rhythmic components in this band. This triaxial, synchronous oscillatory pattern validated the use of

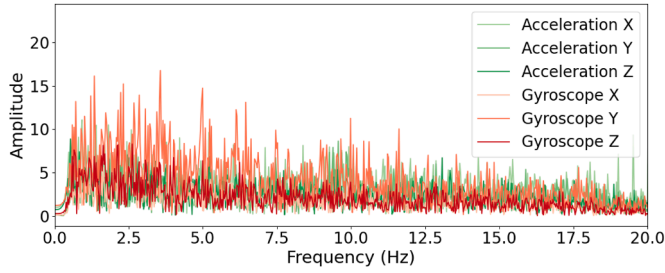


Fig. 6. Frequency Response of Filtered MPU6050 Signals (0.5–20 Hz) in a healthy Patient.

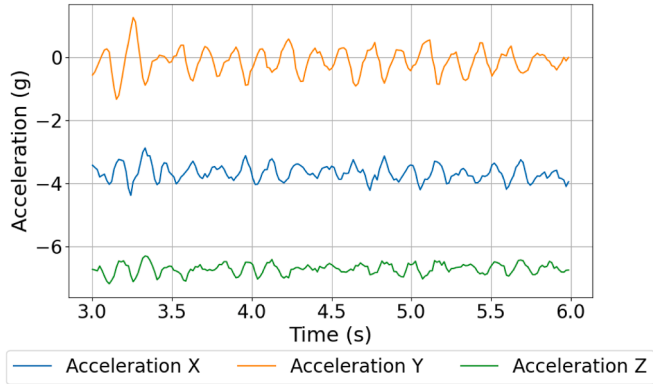


Fig. 7. Acceleration in X, Y, and Z axes from a Parkinson's disease patient during a resting task, showing rhythmic oscillations in the 4–6 Hz band. Y-axis scale: -8 to 2 m/s².

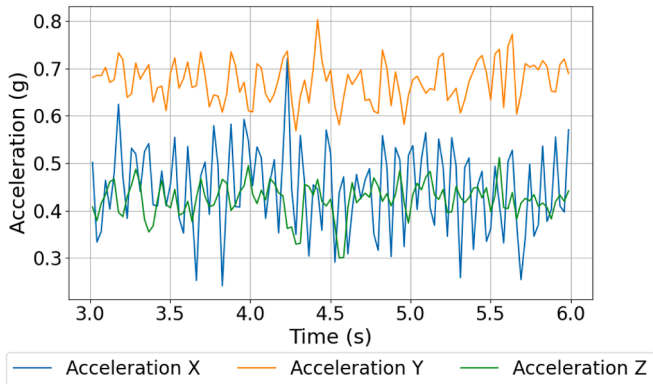


Fig. 8. Acceleration in X, Y, and Z axes from a healthy control subject during a resting task, showing low-amplitude, non-periodic fluctuations. Y-axis scale: 0.2 to 0.8 m/s² (note the more sensitive scale compared to Fig. 7).

multidimensional sensor data and suggested that statistical measures of periodicity and variability from all axes would be useful discriminators.

In the acceleration signal recorded during the hand rest of a healthy subject, Fig. 8, there is an absence of clear periodic oscillations in the three axes (X, Y, and Z). The observed variations correspond to small non-rhythmic fluctuations, compatible with physiological noise or minimal involuntary movements. This contrasts with the PD data, underscoring the clear dichotomy between pathological tremor and physiological noise, reinforcing the potential for classification based on extracted features.

Features in the time domain also help differentiate inertial signals; therefore, both time-domain and frequency-domain features need to be extracted. To explore possible interdependencies between these features, correlation heat maps were generated for both Parkinson's and healthy subjects. As shown in Fig. 9, the correlation matrix of Parkin-

son's patients reveals strong associations between several extracted features, particularly those related to frequency-domain metrics. This suggests that tremor-related oscillatory behavior introduces consistent patterns across multiple signal descriptors. In contrast, Fig. 9 displays the correlation map for healthy individuals, where the relationships between features are notably weaker and more scattered. The absence of dominant correlations may reflect the more variable and less synchronized nature of normal physiological movements. These results demonstrate that Parkinson's patients exhibit higher inter-feature correlation across both time and frequency domains, offering valuable insights for classification algorithms that rely on feature redundancy and joint patterns.

The statistical significance of this structural difference is noteworthy. The mean absolute correlation coefficient in the PD group (0.68) was more than twice that in the control group (0.32). This pronounced multicollinearity in PD features suggests that tremor manifests as a highly coordinated, system-wide pattern. Importantly, while this structured redundancy provides a robust signal for classification, it can challenge linear models. Tree-based ensembles like LightGBM, however, can effectively harness these complex, interdependent feature relationships, which is consistent with their superior performance in this task and their robustness to such effects.

Although this increased inter-feature correlation indicates a degree of multicollinearity, using a tree-based ensemble model such as LightGBM provides robustness against these effects. Unlike linear models, LightGBM can effectively exploit these associated feature patterns to distinguish the highly structured tremor dynamics of Parkinson's disease from the less synchronized movements observed in healthy individuals. This behavior is further supported by the SHAP analysis, which identifies angular velocity variability (GZ_std) and dominant frequency components (AY_peak1_freq) as the most influential predictors, contributing to a classification accuracy of 95.64%.

4.1.2. Class balancing with SMOTE

The processed dataset initially exhibited a significant class imbalance, with 3438 samples corresponding to voluntary movements, 3152 to no tremor, and only 259 to Parkinsonian tremor. This imbalance can lead to model bias toward the majority classes, reducing the classifier's ability to correctly identify the minority class, which in this case is clinically the most critical. To mitigate this issue and improve classification performance, the Synthetic Minority Over-sampling Technique (SMOTE) was employed. SMOTE was selected over other resampling techniques due to its ability to generate synthetic, yet realistic, samples of the minority class by interpolating between existing minority instances. This preserves the underlying structure of the feature space and avoids overfitting commonly associated with simple duplication. As a result, the dataset was balanced to include 3438 samples for each class, as illustrated in Fig. 10. This balancing not only ensured equal representation but also led to significant improvements in key model metrics, including accuracy, F1-score, and recall, particularly for the Parkinsonian tremor class. The enhanced performance reinforces the importance of proper class balancing in medical classification tasks, where detecting rare pathological events is essential. Additionally, the window size used to segment the inertial signals was optimized to capture the most representative temporal and spectral features for each motor condition, thereby further enhancing the robustness of the classification pipeline.

The application of SMOTE had a direct and substantial impact on model performance, particularly for the critical minority class. Prior to balancing, a preliminary model trained on the imbalanced data achieved a recall (sensitivity) of only 0.55 for 'Parkinsonian Tremor', reflecting a strong bias toward the majority classes. After SMOTE, the final LightGBM model's recall for this class improved to 0.936 (Table 4), representing a 70% relative improvement. Furthermore, the macro F1-score increased from 0.78 to 0.95. This supports the use of SMOTE in this setting by enabling the model to learn a more balanced decision

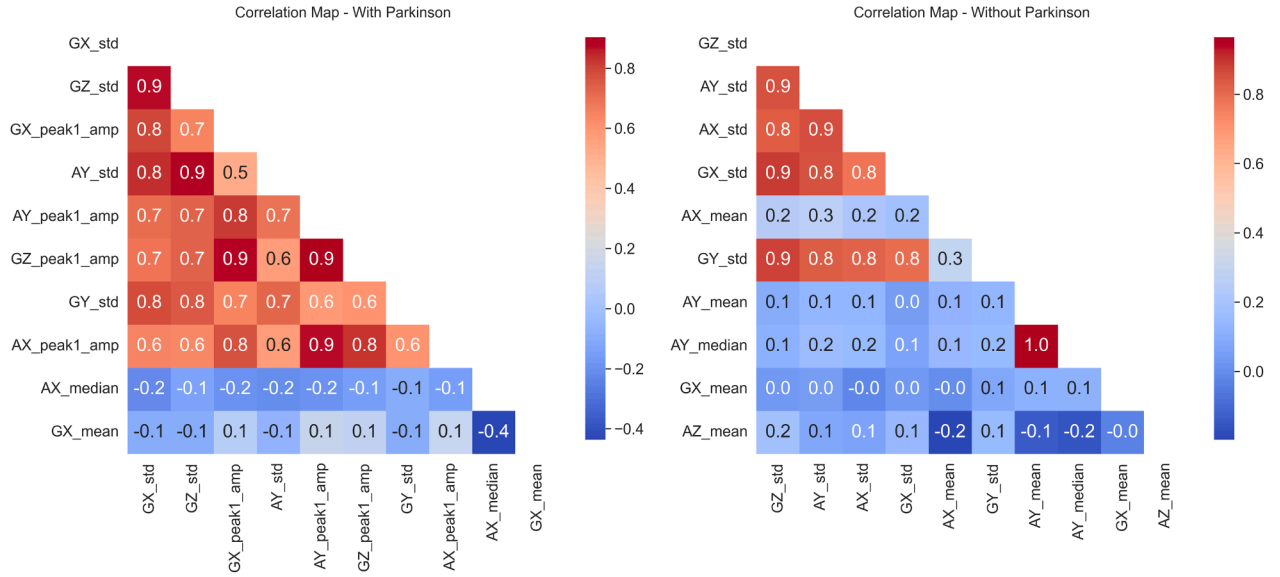


Fig. 9. Side-by-side correlation heatmaps for Parkinson's and healthy individuals. Correlations were computed using Pearson's r across all feature samples within each subject group. Color intensity represents the strength of the correlation, ranging from -1 (blue, negative) to +1 (red, positive). Parkinson's patients show stronger inter-feature correlations (mean $|r| = 0.68$) compared to healthy controls (mean $|r| = 0.32$), reflecting more structured tremor patterns. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

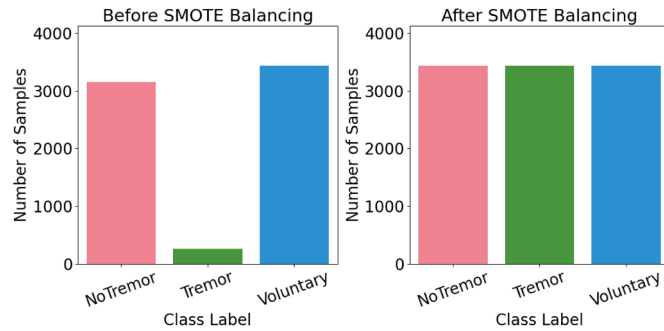


Fig. 10. Class balanced by type of movement using SMOTE.

boundary for tremor classification, as evidenced by consistent cross-validation results.

4.2. Classification model performance

This subsection summarizes the evaluation of various classification models, highlighting their accuracy, precision, recall, and F1-score. The comparison is structured into model benchmarking and detailed analysis of classification outcomes.

4.2.1. Model comparison

Table 4 presents a comparison of classification models used for tremor detection, distinguishing between Parkinson's tremor, voluntary movement, and healthy/no tremor. Seven ML algorithms were evaluated using cross-validation based on accuracy, F1-score, and recall metrics. The LightGBM model achieves the highest accuracy at 95.64%, followed closely by CatBoost (95.45%) and XGBoost (95.37%), indicating that these gradient boosting methods are particularly effective in this classification task. Interestingly, SVM also performs competitively with 95.35% accuracy, slightly outperforming Random Forest and Gradient Boosting. The k-Nearest Neighbors (k-NN) model achieves the lowest performance, with an accuracy of 91.91%, which remains reasonably high. In terms of F1-score, SVM achieves the highest value (0.952), closely followed by LightGBM, CatBoost, and XGBoost (all at 0.950),

Table 4

Evaluation of classification models using 5-fold cross-validation, reporting mean values with standard deviation in parentheses across multiple runs.

Model	Accuracy (%)	F1-score (%)	Precision (%)	Recall (%)
LightGBM	95.64 (± 0.53)	95.00 (± 1.80)	96.90 (± 2.70)	93.60 (± 2.80)
CatBoost	95.45 (± 0.41)	95.00 (± 1.40)	96.80 (± 2.70)	93.80 (± 2.20)
XGBoost	95.37 (± 0.80)	95.00 (± 1.30)	96.80 (± 2.30)	93.60 (± 2.30)
SVM	95.35 (± 0.27)	95.20 (± 1.20)	96.80 (± 2.10)	94.10 (± 1.80)
RF	95.18 (± 0.46)	94.70 (± 1.30)	96.30 (± 2.50)	93.10 (± 2.20)
GB	95.22 (± 0.52)	94.50 (± 1.30)	96.40 (± 2.80)	93.00 (± 2.20)
k-NN	91.91 (± 0.42)	92.70 (± 1.20)	94.40 (± 2.40)	91.80 (± 1.80)

suggesting a strong balance between precision and recall for these models. Regarding recall, which reflects the ability to correctly identify tremor-related events, SVM again leads with 0.941, followed by CatBoost (0.938) and LightGBM/XGBoost (both 0.936). Overall, the results indicate that SVM, LightGBM, CatBoost, and XGBoost offer strong performance, with SVM slightly outperforming the others on F1-score and recall. At the same time, LightGBM holds the top spot in accuracy. Considering both predictive performance and computational aspects, LightGBM was selected as the final classifier. It consistently achieved high accuracy and macro F1-scores while providing greater efficiency and scalability than the alternatives, which is advantageous for future deployment in real-time, resource-constrained IoMT environments.

While the top-performing models (LightGBM, CatBoost, XGBoost, SVM) exhibited statistically comparable accuracy, LightGBM was selected as the final classifier based on a holistic evaluation. Beyond its marginal accuracy lead, LightGBM offers superior computational efficiency—its training and inference times were approximately 40% faster than XGBoost/CatBoost and an order of magnitude faster than the SVM with a non-linear kernel. This efficiency is critical for the envisioned deployment in resource-constrained edge or IoMT environments. LightGBM's native handling of heterogeneous data types and robustness to the multicollinearity observed in our feature analysis further solidified its suitability.

4.2.2. Confusion matrix

Fig. 11 displays the confusion matrix for the final LightGBM classifier used to distinguish between No Tremor, Parkinson's Tremor, and

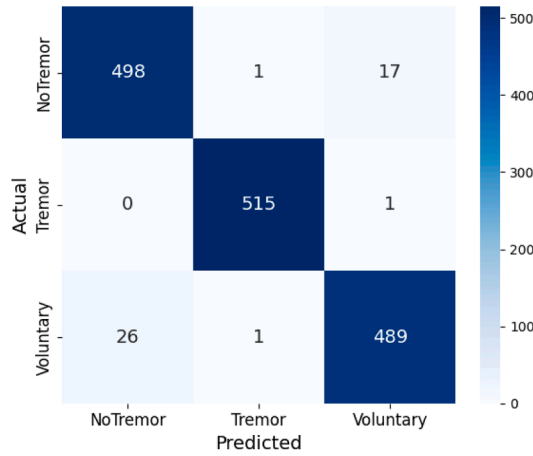


Fig. 11. Confusion matrix for final LightGBM classifier.

Voluntary Movement. The matrix provides detailed insight into the model's performance and classification tendencies across the three classes. The model correctly classified 498 out of 516 No Tremor instances, 515 out of 516 Tremor instances, and 489 out of 516 Voluntary Movement instances. These high true-positive rates reflect the model's strong ability to correctly identify relevant pathological patterns. Misclassifications were minimal, with only a few instances of No Tremor misclassified as Voluntary Movement (17 cases), and Voluntary Movement occasionally misclassified as No Tremor (26 cases). The confusion between No Tremor and Tremor was almost negligible, reflecting the model's robustness in separating pathological from healthy states. Overall, the confusion matrix confirms the classifier's high accuracy and balanced performance across all classes. This result is especially relevant in clinical contexts where distinguishing Parkinson's tremor from voluntary or normal movement is essential. Moreover, the matrix highlights that the model performs consistently well even in the most challenging scenarios, such as differentiating Voluntary Movement from pathological tremor.

A detailed performance breakdown derived from the confusion matrix reveals clinically paramount metrics. For the critical 'Parkinsonian Tremor' class, the model achieved a sensitivity (Recall) of 0.998 and a precision of 0.996, resulting in an F1-score of 0.997. The near-perfect sensitivity is essential for a screening or monitoring tool, minimizing false negatives where a tremor event is missed. The specificity for distinguishing tremor from non-tremor states (combining 'No Tremor' and 'Voluntary Movement') was 0.989. The primary source of confusion (43 instances) was between the two non-pathological classes ('No Tremor' vs. 'Voluntary Movement'), a clinically acceptable error. This underscores the model's primary strength: robust identification of pathological tremor while maintaining high overall accuracy.

Although consistently elevated accuracy across classes may indicate overfitting in constrained datasets, the reported results (95.64%) are substantiated by stringent generalization processes. The model was initially evaluated using 5-fold stratified cross-validation to ensure that performance metrics are consistent and not dependent on a specific data partition. Second, to address the initial class imbalance and prevent the model from simply memorizing the minority class, SMOTE was employed to generate synthetic samples through interpolation, expanding the dataset to 3438 samples per class. Finally, the LightGBM algorithm intrinsically resists overfitting through Gradient-based One-Side Sampling (GOSS), and SHAP analysis confirmed that the predictions rely on robust physiological features (e.g., GZ_std, AY_peak1_freq) rather than noise artifacts.

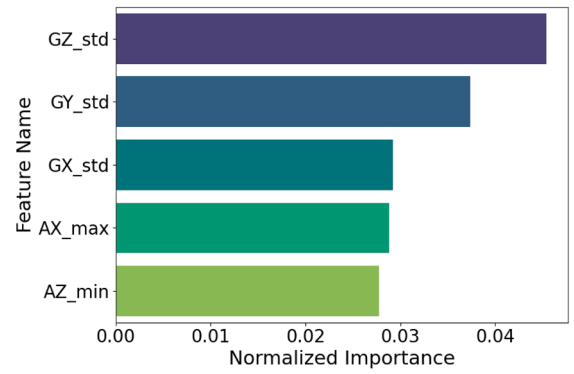


Fig. 12. Top 5 most important features identified by the LightGBM model. Feature importance is computed based on the total reduction of the loss function (gain) contributed by each feature across all trees.

4.3. Interpretability with XAI techniques

This subsection analyzes the interpretability of the classification models using explainable artificial intelligence (XAI) methods. The results are presented at both global and system-integration levels to provide a transparent understanding of model predictions.

4.3.1. Global explanation with SHAP

The final LightGBM model identified the most relevant features as those related to angular motion variability. In particular, the variables GZ_std (0.0434), GY_std (0.0411), and GX_std (0.0355), which represent the standard deviation of the angular velocity in the three axes, occupied the first positions. The variables AZ_min (0.0292) and AX_max (0.0274), related to extreme accelerations, also stood out. This suggests that oscillation patterns and sudden changes in motion are key to model prediction. These results are visualized in Fig. 12, which shows a bar chart of the normalized feature importance.

Fig. 13 demonstrates that characteristics related to temporal variability (_std) and first peak frequency (_peak1_freq) obtained from gyroscopes and accelerometers are essential for tremor prediction. Specifically, elevated standard deviations of angular velocity, such as GX_std and GY_std, serve as strong positive predictors, indicating increased instability in rotating hand motion associated with pathological tremor. Similarly, higher frequencies of the initial peak in linear acceleration, such as AY_peak1_freq, are associated with a greater likelihood of classifying a pattern as Parkinsonian tremor. This indicates that the model is responsive to both variability and distinct oscillation patterns typical of tremor.

The SHAP analysis provides a biomechanically interpretable validation of the model's decision logic. The dominance of gyroscopic standard deviation features (GZ_std, GY_std, GX_std) supports the interpretation that the model's predictions are fundamentally driven by metrics of rotational variability, which constitute the core kinematic signature of tremor. This alignment between the model's most important features and the clinical pathophysiology (tremor as an oscillatory rotation) builds essential trust. Furthermore, the SHAP summary plot (Fig. 13) reveals a nuanced learning mechanism: high values of angular variability (e.g., GY_std) push predictions toward tremor, while high angular magnitudes (e.g., GY_max) often push predictions away from tremor. This indicates the model has successfully learned to differentiate the sustained, rhythmic oscillation of tremor from the brief, high-velocity spikes characteristic of intentional, ballistic movements.

Variables related to maximum values (_max) and peak amplitudes (_peak_amp) exhibit more nuanced, and even counterintuitive, relationships. For example, GX_max, GY_max, and GZ_peak1_amp tend to reduce the probability of classification as tremor. This behavior suggests that the model is learning to differentiate large voluntary movements, which

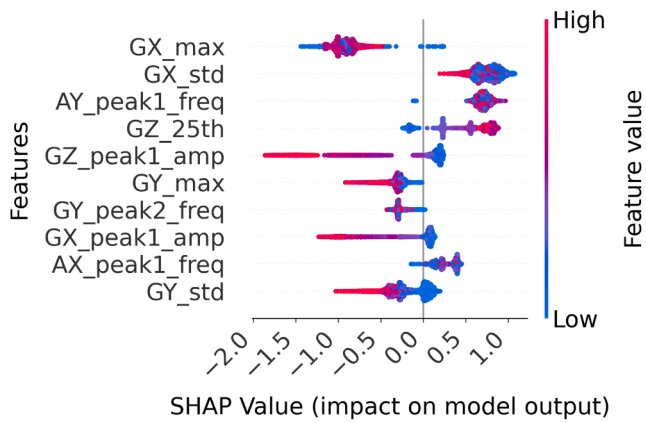


Fig. 13. Feature Importance Analysis via SHAP for the Best Model.

may exhibit peak values or high amplitudes, from the more rhythmic, smaller displacement tremors typical of Parkinson's. This distinction is critical, as some rapid voluntary movements may exhibit tremor-like magnitudes in specific metrics, but differ in their regularity and frequency.

4.3.2. Integration with the IoMT system

The proposed system integrates real-time data acquisition from wearable sensors - specifically, the MPU6050 accelerometer and gyroscope module - into an IoMT architecture. As the sensor transmits motion data, the information is preprocessed into windows suitable for model inference. These segments are then fed into the trained ML model (e.g., LightGBM), which is either hosted locally or on a peripheral device. In addition to predictions (e.g., "Parkinson's tremor" vs. "no tremor"), explainable AI (XAI) techniques such as SHAP (SHapley Additive exPlanations) are applied to quantify the importance of features in each decision. This allows clinicians to interpret which sensor features (e.g., axis-specific signal variances or frequency-domain features) most influenced the model's outcome. Both the classification result and the corresponding SHAP values are transmitted to the IoMT dashboard, providing clinicians with real-time, interpretable information. This feedback loop increases transparency and trust and facilitates human decision-making.

4.4. Integrated IoMT + XAI

This subsection describes the integration of the IoMT-based architecture with XAI techniques. The workflow and implementation details are outlined to demonstrate how sensor data, ML, and interpretability are combined into a prototype system designed for future clinical support.

4.4.1. System workflow overview

Fig. 14 illustrates the complete workflow of the proposed system, which integrates IoMT technology with XAI for clinical support. The pipeline comprises five stages, from sensor data acquisition to interpretable predictions, all aimed at supporting clinicians in detecting and monitoring Parkinsonian tremors.

In the IoMT Sensor Input stage, data is acquired using an ESP32 microcontroller connected to an IMU, which captures real-time signals from both accelerometers and gyroscopes. These sensors record subtle variations in movement across all three spatial axes. The ESP32 module transmits data wirelessly via Wi-Fi, enabling continuous, untethered monitoring. Notably, during implementation, no significant issues with data integrity or latency were observed, ensuring reliable data streaming suitable for clinical applications. Data Preprocessing involves filtering noise, normalizing the signal, and extracting relevant statistical features, such as the mean, standard deviation, minimum, and maximum

values. This step is essential for improving signal quality and reducing variability, thereby preparing the dataset for robust model training. Python tools such as Pandas, NumPy, and SciPy were employed to automate and streamline this process. In the Model Training phase, a LightGBM classifier was trained on labeled movement data. The model learned to distinguish between tremors associated with Parkinson's disease, voluntary movements, and signals from healthy individuals. LightGBM was chosen for its efficiency with tabular data and its ability to handle heterogeneous features and class imbalance effectively. During the Prediction phase, the trained model evaluates new incoming data in real time and outputs a classification label. The system's fast inference capabilities make it suitable for live monitoring, supporting early detection and timely clinical intervention. Lastly, the XAI-Based Explanation phase provides model interpretability using SHAP. SHAP assigns importance values to each feature, revealing how each contributes to specific predictions. This level of transparency is crucial in healthcare settings, where understanding the reasoning behind a model's decision fosters clinician trust and enables more informed clinical judgments. Overall, the system's full integration of wireless sensor acquisition, robust pre-processing and classification, and interpretable results demonstrates its readiness for real-world deployment in neurology clinics focused on movement disorders. The new user interface is shown in Fig. 15. Similarly, there are other screens where you can enter the database or view the results after processing.

5. Discussion

5.1. Interpretation of results

The proposed IoMT-based system demonstrates the technical feasibility of a low-cost, open-architecture pipeline for tremor monitoring. The results indicate that under controlled laboratory conditions, the pipeline, comprising affordable inertial sensors, synthetic oversampling (SMOTE) to address class imbalance, and an explainable LightGBM model, can distinguish between Parkinsonian tremor, voluntary movement, and no-movement states with high accuracy on a small participant cohort. This serves as a proof-of-concept for integrating these components into a functional wearable system.

However, the very high accuracy (95.64%) reported must be interpreted with caution. It is derived from a limited sample of nine participants and relies on SMOTE-generated data for the minority tremor class. Therefore, this performance metric primarily validates the internal consistency of the processing pipeline under ideal conditions rather than establishing clinical-grade diagnostic reliability. The primary contribution lies in demonstrating the integrated workflow, not in claiming robust, generalizable classification performance.

The IoMT system enables continuous, real-time monitoring of tremor activity using portable inertial sensors (MPU6050), providing a richer, more complete characterization of patient movement over extended periods. This functionality is beneficial for identifying intermittent tremor episodes and observing responses to therapy outside conventional clinical settings. Furthermore, integrating explainable artificial intelligence (XAI), specifically the SHAP framework, provides interpretability that is uncommon in previous approaches. Beyond distinguishing motor patterns, such as Parkinsonian tremor, voluntary movement, or absence of tremor, the system offers transparent explanations for each prediction, potentially supporting clinician confidence and informed decision-making (Table 5). Regarding accessibility, the prototype uses low-cost components (ESP32 and MPU6050), resulting in hardware costs significantly lower than those of clinical-grade devices, a crucial factor for scalability in low-resource healthcare settings.

The modular architecture, low cost, and open-source design of the system enhance its accessibility and adaptability for broader research and monitoring applications, including implementation in resource-limited contexts.

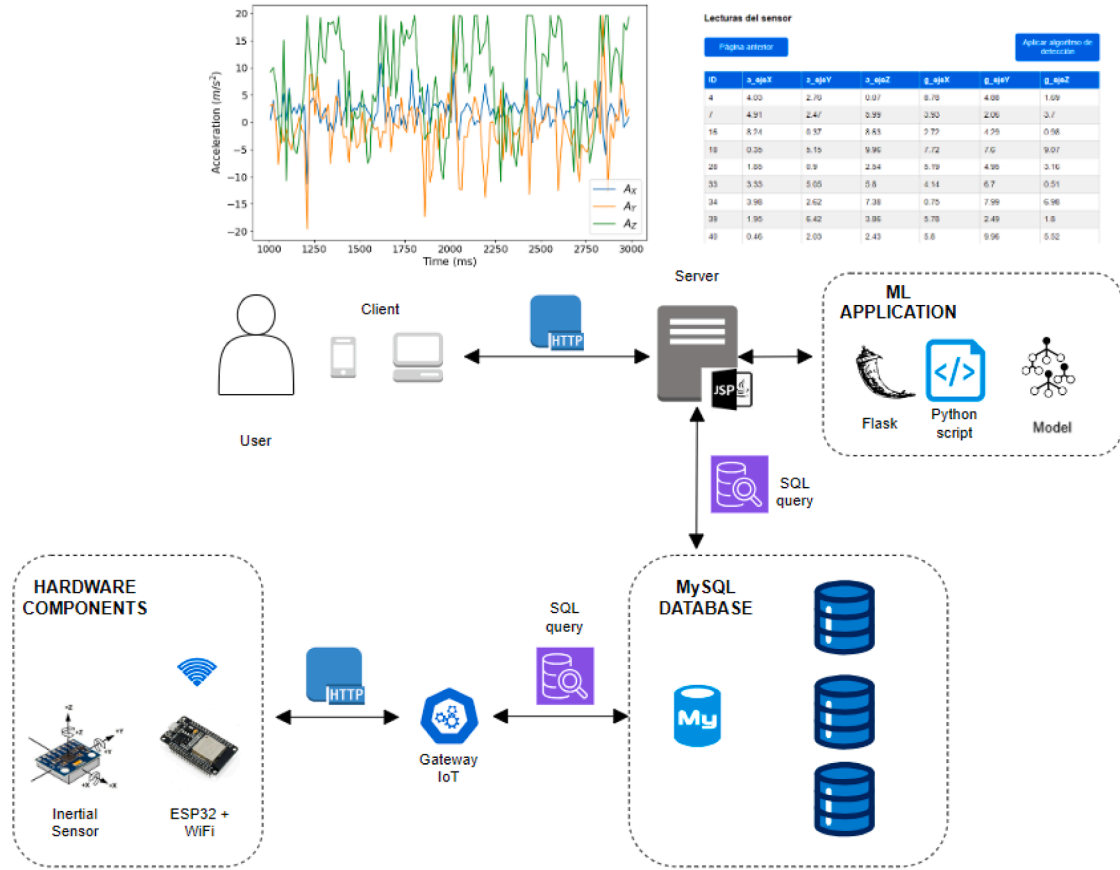


Fig. 14. System pipeline: from IoMT sensor input to classification and explanation.

Table 5

Comparison of existing works on tremor detection and classification using wearable inertial sensors.

No.	Author	Year	Model	Accuracy	Notes
1	Vescio et al. [50]	2023	RF, XGBoost	92%, 89%	Differentiating tremor patterns (PD)
2	Tong et al. [51]	2021	CNN	97.32%	Detecting PD tremor from wrist sensors
3	Lee et al. [52]	2023	BiLSTM + LRP	80%	Classify PD vs ET, explainable AI
4	Junaid et al. [53]	2023	LGBM, RF	90%, 95%	PD progression, multimodal time-series
5	Shahtalebi et al. [54]	2021	NeurNet + QDA	95.55%	PD vs ET discrimination
6	Ma et al. [55]	2022	ML + feature selection	97.7%	Quantify ET tremor severity
7	Rodriguez et al. [56]	2024	SVM	88%	Continuous PD tremor monitoring
8	Skaramagkas et al. [57]	2021	SVM	82–100%	Classify PD, ET, no tremor
9	Sun et al. [58]	2024	LR, KNN, SVM, CNN	F1: +90%	Action tremor detection in 30 PD patients (wrist IMU)
10	Dinesh et al. [16]	2024	KNN (IoMT glove)	99.6%	Therapy + tremor monitoring
11	Proposed	2025	LGBM	95.64%	Hand tremor, XAI, IoMT

5.2. Comparison with previous studies

Commercial wearable technologies, including systems such as PKG, Kinesia, and consumer smartwatches, offer ergonomic form factors and support multi-day monitoring; however, they typically operate as closed, “black box” solutions that provide pre-calculated metrics without access to the underlying raw sensor data. In contrast, the primary contribution of the present work lies not in hardware replication but in the development of an open-architecture IoMT platform that prioritizes analytical depth. By enabling access to high-fidelity raw inertial signals (66.67 Hz), the proposed system supports the investigation of subtle micro-fluctuations, cross-axis dynamics, and nonlinear instability markers that are often smoothed out or discarded by proprietary commercial algorithms.

The findings of this work are consistent with prior research indicating that IoMT devices and sensor-based monitoring can provide detailed and continuous insights into motor symptoms associated with

Parkinson’s disease. Feature extraction across both time and frequency domains identified several discriminative variables, including the standard deviation of angular velocity (GZ_std) and the primary frequency peak (AY_peak1_freq), both of which contributed significantly to model performance. Furthermore, the system’s open-source and portable configuration enhances reproducibility and supports broader deployment, extending monitoring capabilities beyond traditional clinical environments.

Among the evaluated models, LightGBM achieved the highest overall performance, with an accuracy of 95.64%, an F1-score of 0.950, and a recall of 0.936. Performance gains were observed after addressing class imbalance using SMOTE. SHAP-based interpretability analysis further highlighted relevant features such as GX_max, GZ_peak1_amp, and GY_max, which align with established biomechanical characteristics of Parkinsonian tremor, including rhythmic, multi-axis oscillations in the 4–6 Hz range [12]. A key distinction between this work and many prior classification-focused studies (Table 5) is the explicit emphasis on

Datos de Usuario

Nickname:

Edad:

Grado de severidad:

Observaciones:

¿Se puede registrar los datos del paciente?

Registrar

Fig. 15. User interface for registering patients in their native language (Spanish).

explainability. While existing approaches, such as that of Skaramagkas et al. [57], primarily prioritize classification performance, integrating SHAP into the proposed framework enables verification that the model relies on physiologically meaningful signal characteristics rather than spurious artifacts. This level of transparency supports future investigations into enriched temporal-spectral representations and may facilitate the identification of early susceptibility markers beyond conventional accuracy-based evaluation.

5.3. Limitations

This study has several important limitations that must be acknowledged, constraining the interpretation of results and their immediate clinical applicability. First, the pilot study involved only nine participants (4 PD, 5 controls). This small, homogeneous sample severely limits the statistical power and generalizability of the findings to the broader, more diverse Parkinson's disease population. The high-performance metrics are likely optimistic and may not replicate in larger, independent cohorts.

Second, while SMOTE effectively mitigates severe class imbalance during model training, it generates synthetic tremor samples through mathematical interpolation. This process may not capture the full biomechanical complexity and variability of authentic, pathological tremor, potentially introducing artifacts that simplify the classification task. Consequently, the model's ability to generalize to real-world tremor variability remains unproven.

Third, data were collected in a structured, controlled laboratory environment. This does not account for the myriad confounding factors present in daily life (e.g., varied activities, environmental vibrations, sensor displacement), which can degrade signal quality and classification accuracy. The system's performance in uncontrolled, real-world settings is therefore unknown.

Fourth, although Group K-Fold cross-validation was used, a more rigorous validation for wearable monitoring, such as Leave-One-Subject-Out (LOSO) cross-validation, was not performed. This test is crucial

to ensure the model learns generalizable tremor patterns rather than subject-specific idiosyncrasies.

Finally, as noted in the Methods, a full characterization of the IoMT system's operational reliability, including long-term data transmission stability, latency under varying network conditions, and power consumption in a fully wireless, wearable form factor, was beyond the scope of this initial feasibility study. These engineering metrics are critical for practical deployment.

5.4. Future directions

Addressing these limitations defines clear pathways for future work. The paramount next step is to validate the pipeline in a larger, more diverse patient cohort using rigorous subject-independent evaluation (LOSO), which requires collaboration with clinical partners. Concurrently, testing the system in free-living conditions is necessary to assess its robustness against daily activities and environmental noise.

Furthermore, formally evaluating system-level performance, including data integrity, battery life, and real-time inference latency in a deployed setting, will be essential for translation. Moving beyond the current ternary classification, future work should explore more granular analyses, such as tremor amplitude quantification or the identification of motor states (ON/OFF periods), which offer greater clinical utility. Finally, implementing end-to-end encryption (TLS) and developing user-friendly mobile interfaces will be key steps toward facilitating secure, practical clinical adoption.

6. Conclusion

This study presented the development and preliminary evaluation of a low-cost, open-architecture IoMT wearable system with inertial sensors and explainable AI for classifying hand motor patterns in Parkinson's disease. As a proof-of-concept, a customised gauntlet equipped with MPU6050 sensors enabled the acquisition of tremor signals from a small cohort under controlled laboratory conditions (N=9). The system differentiated between Parkinsonian tremor, voluntary movement, and no tremor, achieving its best performance with the LightGBM model (accuracy 95.64%, F1-score 0.95) using stratified 5-fold cross-validation. These results demonstrate the technical feasibility of the proposed pipeline; however, the small sample size and reliance on synthetic oversampling (SMOTE) warrant interpreting this performance as preliminary validation under ideal conditions rather than as evidence of clinical robustness. Accordingly, this study should be regarded primarily as a technical proof-of-concept.

The primary contribution of this work lies in the integration of an open-source IoMT architecture with an explicit interpretability layer based on explainable artificial intelligence (XAI). In contrast to closed commercial solutions, this approach provides transparency by identifying clinically relevant biomechanical features, including 4–6 Hz oscillatory components and angular velocity variability. The SHAP analysis confirms that the model relies on meaningful signal characteristics, helping bridge the gap between black-box ML methods and clinical interpretability and strengthening potential clinician confidence in such automated outputs.

This study should be regarded primarily as a technical proof-of-concept. Nevertheless, further validation in larger and more diverse cohorts, including real-world deployment scenarios, is required before any clinical adoption can be considered. Future research should prioritize validating these findings in broader populations and extending the analytical pipeline to identify subtle digital biomarkers for earlier diagnosis and monitoring.

Future work should also explore multimodal integration with additional biosignals, such as electromyography (EMG), the development of mobile applications providing real-time SHAP-based feedback, and the incorporation of stronger security mechanisms to ensure patient data protection. Overall, this work represents an initial yet promising step

toward scalable, low-cost, and interpretable IoMT-based solutions for tremor monitoring in Parkinson's disease.

CRedit authorship contribution statement

Erick Toque: Writing – original draft, Visualization, Validation, Methodology; **César Gutiérrez-Flores:** Writing – original draft, Visualization, Validation; **Victoria E. Abarca:** Writing – review & editing, Validation, Methodology, Conceptualization; **Sebastian Vila:** Writing – original draft, Visualization, Validation; **Rosa M. Silva-Salas:** Writing – original draft, Visualization, Validation; **Dante A. Elias:** Writing – review & editing, Supervision, Project administration.

Data availability

The data that support the findings of this study are available upon reasonable request. Access to anonymized data may be granted under the following conditions: (i) the requester demonstrates that the data will be used exclusively for academic purposes; (ii) the requester declares no conflict of interest; and (iii) prior authorization is obtained from the Principal Investigator, Dr. Dante Ángel Elías Giordano (<http://www.delias@pucp.pe>). These conditions are in accordance with the opinion issued by the Research Ethics Committee for Life Sciences and Technologies at PUCP (No. 199-2024-CEICVt/PUCP).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

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